Machine Learning Model Outcomes

Executive summary report for TikTok prepared by the TikTok data team

Overview

This project builds a machine learning model to automatically classify TikTok videos as either claims or opinions.
The goal is to help prioritize user-reported videos for human review.

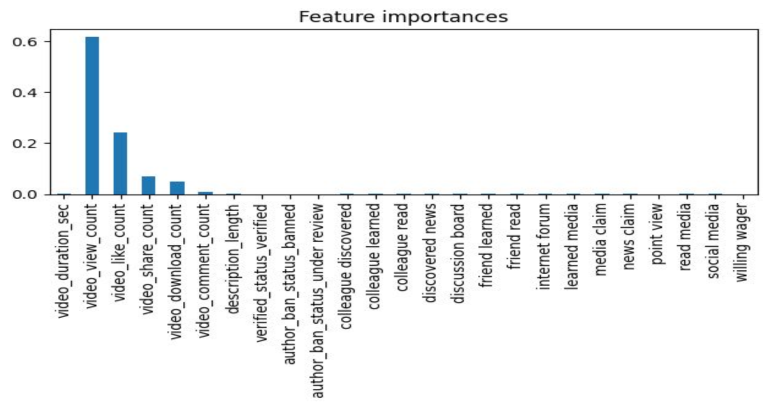
Problem

TikTok receives a high volume of reported videos that require manual review.
Manually checking every video is slow, costly, and inefficient.
Videos making claims require faster moderation than personal opinions to prevent the spread of misinformation.

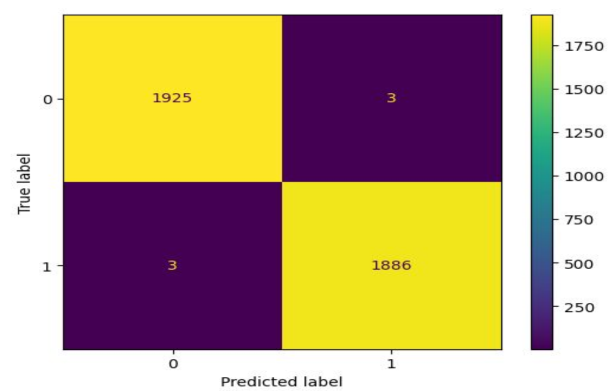
Solution

We built and evaluated tree-based machine learning models using engagement metrics and video details.
The **Random Forest** model was selected as the champion model because it achieved near-perfect accuracy (~99.7%) and high recall, making only 10 errors across 3,817 validation samples.

Details



Confusion matrix for the champion RF model on test holdout data shows only five misclassified samples out of 3,817.



Key Drivers: Video view counts (~60%) and like counts (~25%) were the strongest indicators for predicting claims versus opinions.

Model Performance: Random Forest outperformed XGBoost, capturing more claim videos accurately with fewer false negatives.
Safety Net: Because the model prioritizes recall, almost all claim videos are successfully flagged for moderation.

Next Steps

Deploy to Production: Integrate the Random Forest model into the moderation pipeline to automatically triage incoming video reports.
Feature Expansion: Add NLP-based features (like sentiment analysis or text embeddings from transcripts) to further improve edge-case performance.
Monitor Performance: Continuously track prediction accuracy on live data to ensure performance remains stable over time.